Neopixel Shield
PULSAR FACTOR
Mfr. Part #: UE0118

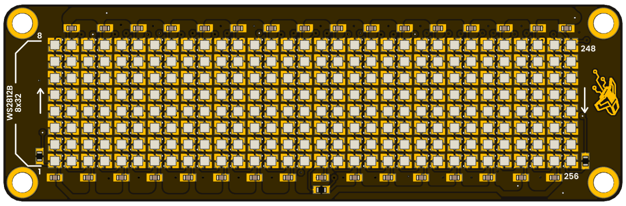
Product Reference Manual (V1.0)

Description

The DevLab 8×32 RGB LED Matrix Module is a compact, addressable LED display designed as an integration-ready visual component for embedded and interactive systems. The module implements a fixed 8-row by 32-column pixel array using smart RGB LEDs with integrated control circuitry, enabling precise color representation and uniform behavior across the entire matrix.

Rather than functioning as a standalone display, the matrix is intended to operate as a hardware surface layer within larger platforms, shields, or modular assemblies. Its electrical interfaces, mechanical layout, and expansion headers are optimized for reuse, scalability, and interoperability across multiple product families.

The module follows the DevLab form factor philosophy, emphasizing simplicity, standardized connectivity, and compatibility with existing controller ecosystems. Implemented as a shield-style board, it supports Pulsar-format controllers and provides alignment with XIAO-class modules, enabling compact and interactive applications without requiring custom display hardware.



Key Features

- **Pulsar Shield form factor**
- **8×32 RGB matrix** (256 individually addressable LEDs)
- Smart RGB LEDs (WS2xxx family) with integrated drivers
- **24-bit color depth** per pixel
- Single-wire digital data interface (800 kHz class)
- **Dual data channels** for flexible routing and cascading
- JST connectors and standard pin headers
- **Matrix Joint expansion interface** for panel extension
- Nominal **5 V operating voltage**
- High brightness with wide viewing angle
- Designed for stacked and modular systems

Hardware Features

- Fixed 8×32 LED grid with deterministic pixel mapping
- Integrated driver per LED (no external controller IC required)
- Dual data input/output paths (DIN/DOUT pairs)
- 4-pin header for power and data access
- JST connectors for daisy-chain configurations
- Expansion interface supporting multi-panel assemblies
- Rigid PCB with defined mounting points
- Electrical characteristics aligned with Pulsar bus power domains

Software Support

- Compatible with Arduino-based environments
- Supported by common addressable RGB LED libraries
- MicroPython compatibility
- Firmware-agnostic hardware design
- Suitable for custom animation engines and UI frameworks

Applications

- Pulsar-based interactive systems
- Modular RGB display assemblies
- Visual feedback and status panels
- UX/UI lighting prototypes
- Educational and experimental platforms
- Decorative and ambient lighting systems

CONTENTS

1 The Board

1.1 Accessories

1 x QWIIC 4-pin JST 1.0 mm Harness

2 Ratings

2.1 Recommended Operating Conditions

Symbol	Description	Min	Typ	Max	Unit
VDD	Operating supply voltage	3.3	-	5.5	V
VIH	Data input high voltage	0.7×VDD	-	VDD	V
VIL	Data input low voltage	0	-	0.3×VDD	V
fDATA	Data transmission rate	-	800	-	kbps
ILED	Current per color channel	-	2	-	mA
IPIXEL	Current per pixel, RGB full white	-	6	-	mA
IMATRIX	Total current, 256 LEDs full white	-	960	-	mA
TA	Operating ambient temperature	-20	25	85	°C
CIN	Recommended bulk input capacitance	10	100	-	µF
CD	Local decoupling capacitance per section/rail	0.1	1	-	µF

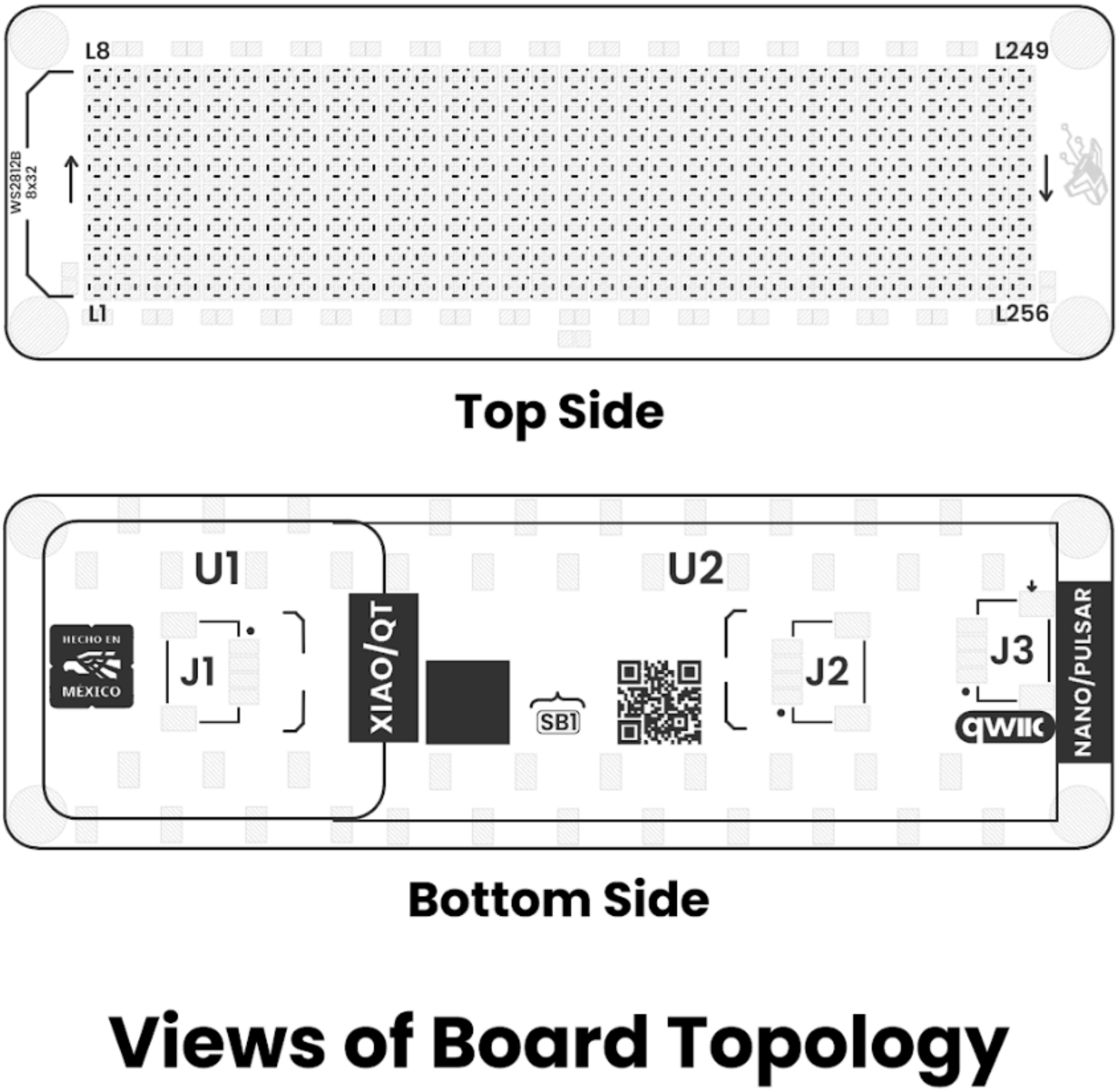
3 Functional Overview

As a **Pulsar Shield**, the DevLab 8×32 RGB LED Matrix serves as a **dedicated visual extension** for compatible controller boards. Its function is to provide a **structured, addressable pixel surface** that integrates mechanically and electrically into Pulsar systems without additional interface hardware.

The shield abstracts low-level LED driving through integrated pixel controllers, allowing the host system to interact with the matrix as a continuous pixel array. Dual data paths and expansion interfaces enable multiple shields or panels to be combined into larger displays while maintaining signal integrity and alignment.

By supporting both **Pulsar and XIAO ecosystems**, the module enables scalable visual designs across different system sizes, from compact interactive nodes to larger tiled display assemblies. The product definition focuses on **integration and compatibility**, leaving visual behavior and control strategy fully determined by the host firmware.

3.2 Board Topology



Views of Module Topology

Table 3.2.1 - Components Overview

Ref.	Description
L1–L256	Addressable RGB LEDs, WS2812B-compatible, 1.0 × 1.0 mm package, SMD 1010.
U1	Header connectors for inserting a Pulsar or Nano-format controller board.
U2	Header connectors for inserting a XIAO/QT-format controller board.

J1	WS2812B data output connector, JST 1.0 mm pitch, for external WS2812B expansion from Matrix 1 and Matrix 2.
J2	WS2812B data input connector, JST 1.0 mm pitch, for Matrix 1 and Matrix 2.
J3	QWIIC connector, JST 1.0 mm pitch, for I²C interface.
SB1	Solder bridge/jumper pad used to join both LED matrices so they can be controlled with a single data signal.

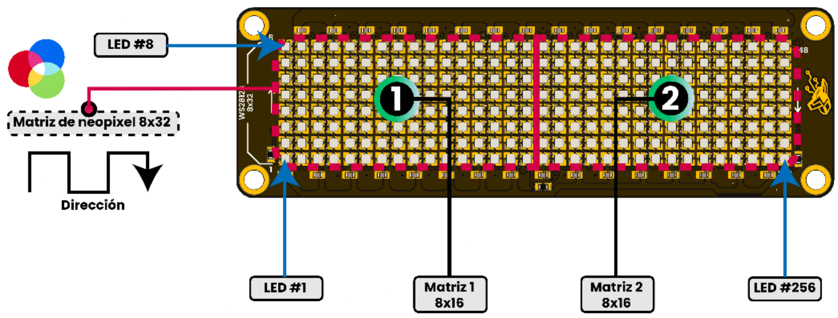
4 Connectors & Pinouts

4.1 General Pinout

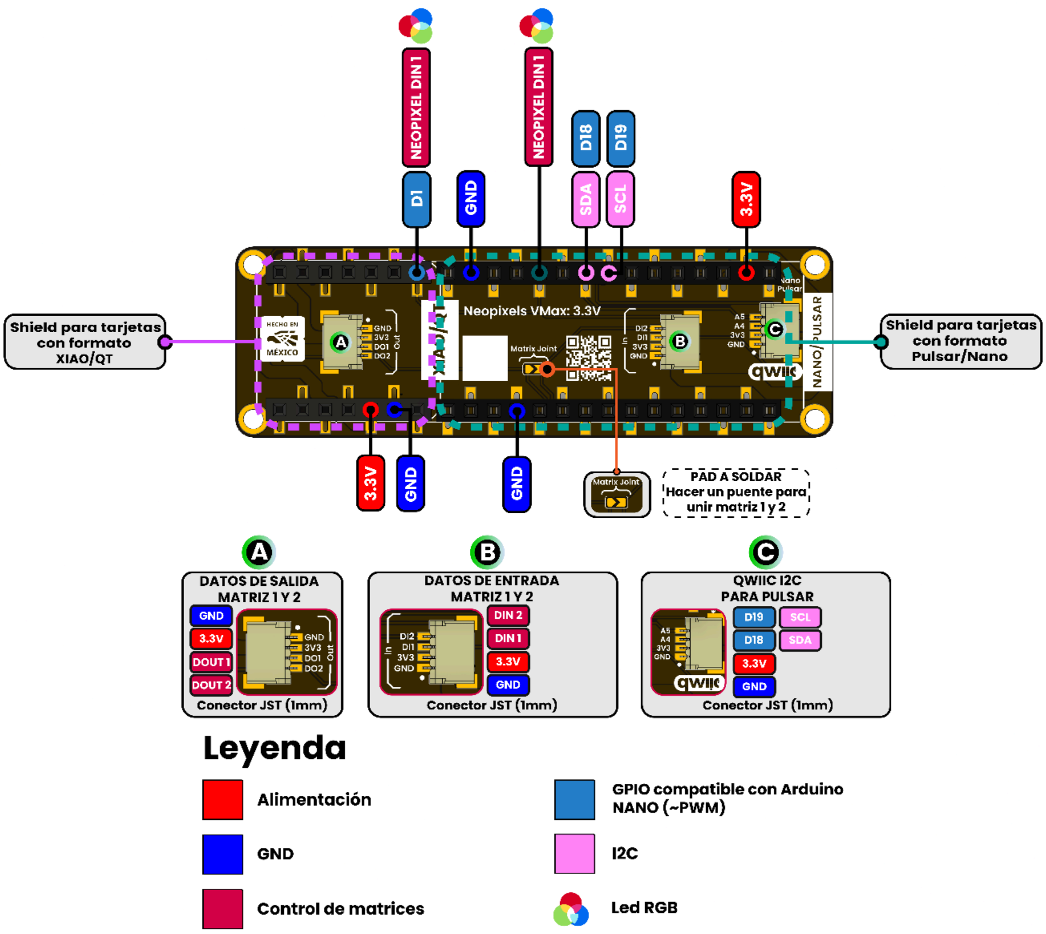
PINOUT

Pulsar: Neopixel Shield

Vista superior



Vista inferior



4.2 Pinout General Description

Pin Label	Function	Notes
+3V3	Power Supply	3.3 V power rail for the shield and external interfaces.
GND	Ground	Common ground reference shared by controller boards, LED matrix and expansion connectors.
SDA	I²C SDA (A4/D18)	Serial Data Line available through the QWIIC connector.
SCL	I²C SCL (A5/D19)	Serial Clock Line available through the QWIIC connector.
DI1	Matrix Data Input 1	Data input for LED Matrix 1 (LED #1 to LED #128). Available on connector J2.
DI2	Matrix Data Input 2	Data input for LED Matrix 2 (LED #129 to LED #256). Available on connector J2.
DO1	Matrix Data Output 1	Data output from Matrix 1 for cascading additional WS2812B devices. Available on connector J1.
DO2	Matrix Data Output 2	Data output from Matrix 2 for cascading additional WS2812B devices. Available on connector J1.
SB1	Solder Bridge	Optional solder jumper used to electrically connect Matrix 1 and Matrix 2 for single-signal operation (**DI1**).

5 Board Operation

5.1 Getting Started with Arduino IDE

The DevLab 8×32 RGB LED Matrix is designed to operate with Arduino-compatible development boards through a single-wire digital interface compatible with WS2812-class LEDs. No additional LED driver ICs are required, as each pixel contains an integrated controller responsible for color generation and signal forwarding.

To begin using the module, connect the controller board to the matrix power and data interfaces. The host microcontroller is responsible for generating the LED data stream and updating the display contents.

Required Components

- DevLab 8×32 RGB LED Matrix
- PULSAR C6
- USB cable
- Adafruit NeoPixel library

Basic Connections

Matrix Signal	Controller
VCC	5 V
GND	GND
DI1	Digital Output
DI2	Optional Digital Output

A common ground connection between the controller and the matrix must always be present.

Example Code

```
#include <Adafruit_NeoPixel.h>

#define LED_PIN    8
#define NUM_LEDS   256

Adafruit_NeoPixel matrix(NUM_LEDS, LED_PIN, NEO_GRB + NEO_KHZ800);

void setup() {
```

```
  matrix.begin();

  matrix.clear();


  matrix.fill(matrix.Color(255, 0, 0));

  matrix.show();

}


void loop() {

}
```

After uploading the sketch, all LEDs should illuminate red, confirming proper communication between the controller and the matrix.

5.2 Pixel Organization

The module contains 256 individually addressable RGB LEDs arranged as an 8-row by 32-column matrix.

Each LED can be controlled independently, allowing the creation of:

- Static graphics
- Text rendering
- Animations
- User interface indicators
- Real-time visualization effects

The host application may treat the display either as a linear LED strip or as a two-dimensional pixel matrix depending on the software framework being used.

5.3 Expansion and Cascading

The module includes dedicated input and output connectors for WS2812-compatible signal routing.

Additional LED matrices or LED strips may be connected to the output interfaces, allowing larger display surfaces to be created without modifying the hardware design.

Typical expansion scenarios include:

- Multi-panel information displays
- Animated lighting installations
- Educational demonstrations
- Interactive exhibits
- Visual monitoring systems

When cascading multiple modules, the host firmware should account for the increased LED count and memory requirements.

5.4 Brightness and Power Considerations

The power consumption of the matrix depends on the number of illuminated LEDs and the selected brightness level.

For most applications, it is recommended to limit global brightness through software to reduce power consumption and thermal load.

Typical recommendations:

- Development and indoor use: 20–50% brightness
- Demonstration systems: 50–80% brightness
- Maximum brightness only when sufficient power capacity is available

Large installations should use an external regulated 5 V power source capable of supplying the required LED current.

5.5 Typical Software Architectures

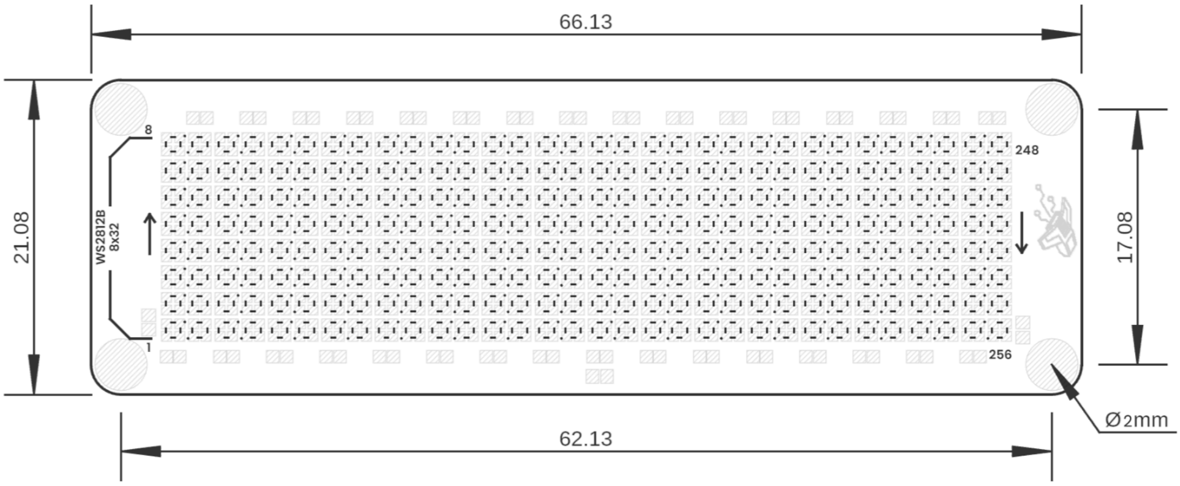
The module is hardware-agnostic and may be integrated into a wide range of software environments.

Common implementations include:

- Arduino and PlatformIO projects
- MicroPython applications
- Real-time sensor visualization
- IoT dashboards
- Educational programming platforms
- Custom graphics engines

The matrix hardware only provides the pixel surface. Animation behavior, graphical rendering, communication protocols, and user interaction logic are entirely defined by the host application.

6 Mechanical Information



Board Dimensions in Millimeters

Mechanical dimensions in millimeters

7 Company Information

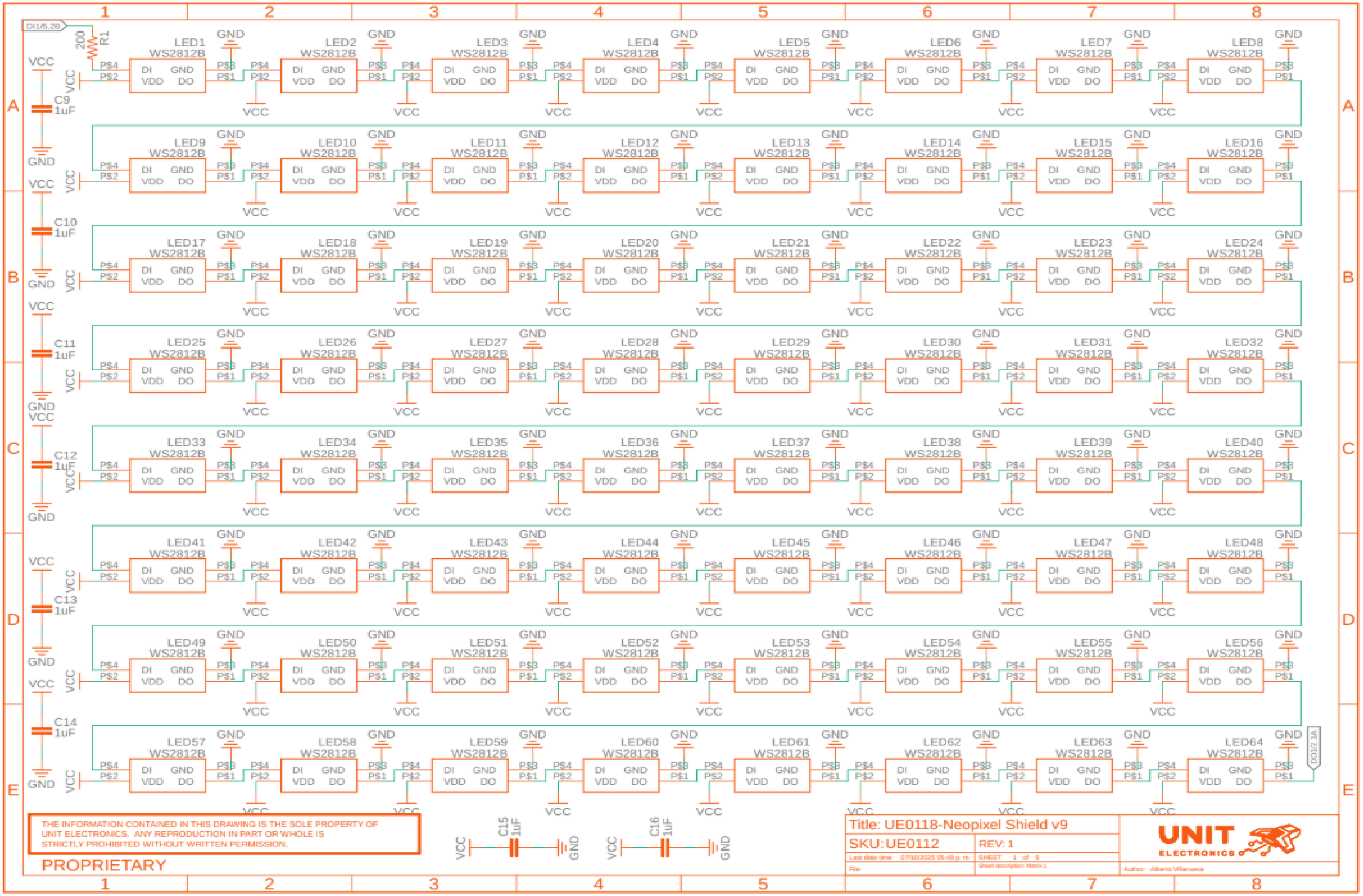
Company name	UNIT Electronics
Company website	https://uelectronics.com/
Company Address	Salvador 19, Cuauhtémoc, 06000 Mexico City, CDMX

8 Reference Documentation

Ref	**Link**
Documentation	https://wiki.uelectronics.com/wiki/unit_pulsar_neopixel_shield
Getting Started Guide	https://github.com/UNIT-Electronics-MX/unit_pulsar_neopixel_shield
Thonny IDE	https://thonny.org/
Arduino IDE	https://www.arduino.cc/en/software
Visual Studio Code	https://code.visualstudio.com/download

9 Appendix

9.1 Schematic (https://github.com/UNIT-Electronics-MX/unit_pulsar_neopixel_shield/blob/main/hardware/unit_sch_v_1_0_0_pulsar_neopixel.pdf)



LED1 WS2812B
LED64 WS2812B
DI GND VDD DO
VCC
GND
C9 1uF
C10 1uF
C11 1uF
C12 1uF
C13 1uF
C14 1uF
C15 1uF
C16 1uF
R1 200
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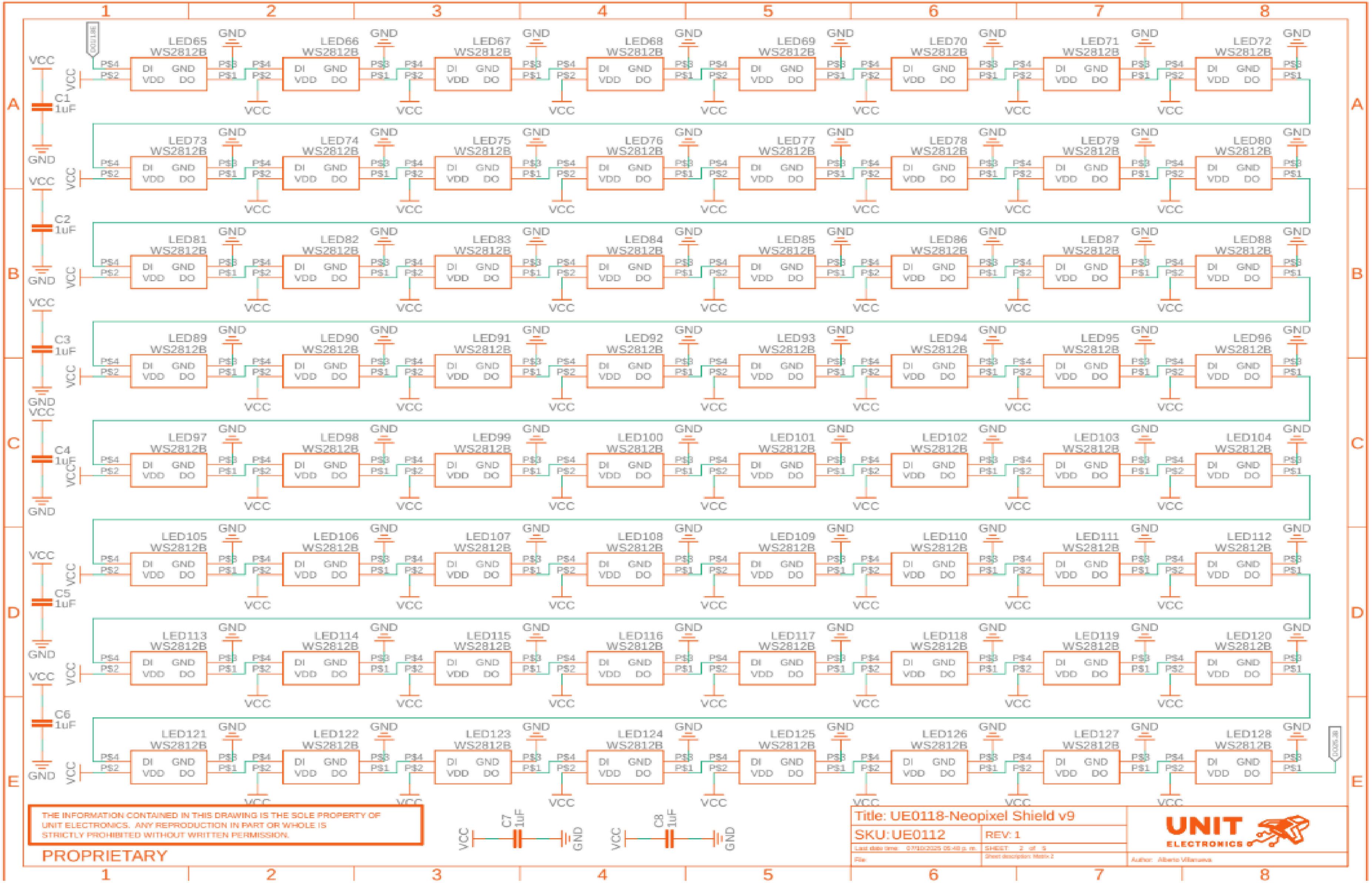
PROPRIETARY
Title: UE0118-Neopixel Shield v9
SKU: UE0112
REV: 1


LED65 WS2812B
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LED67 WS2812B
LED68 WS2812B
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LED73 WS2812B
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LED125 WS2812B
LED126 WS2812B
LED127 WS2812B
LED128 WS2812B
C1 1uF
C2 1uF
C3 1uF
C4 1uF
C5 1uF
C6 1uF
C7 1uF
C8 1uF
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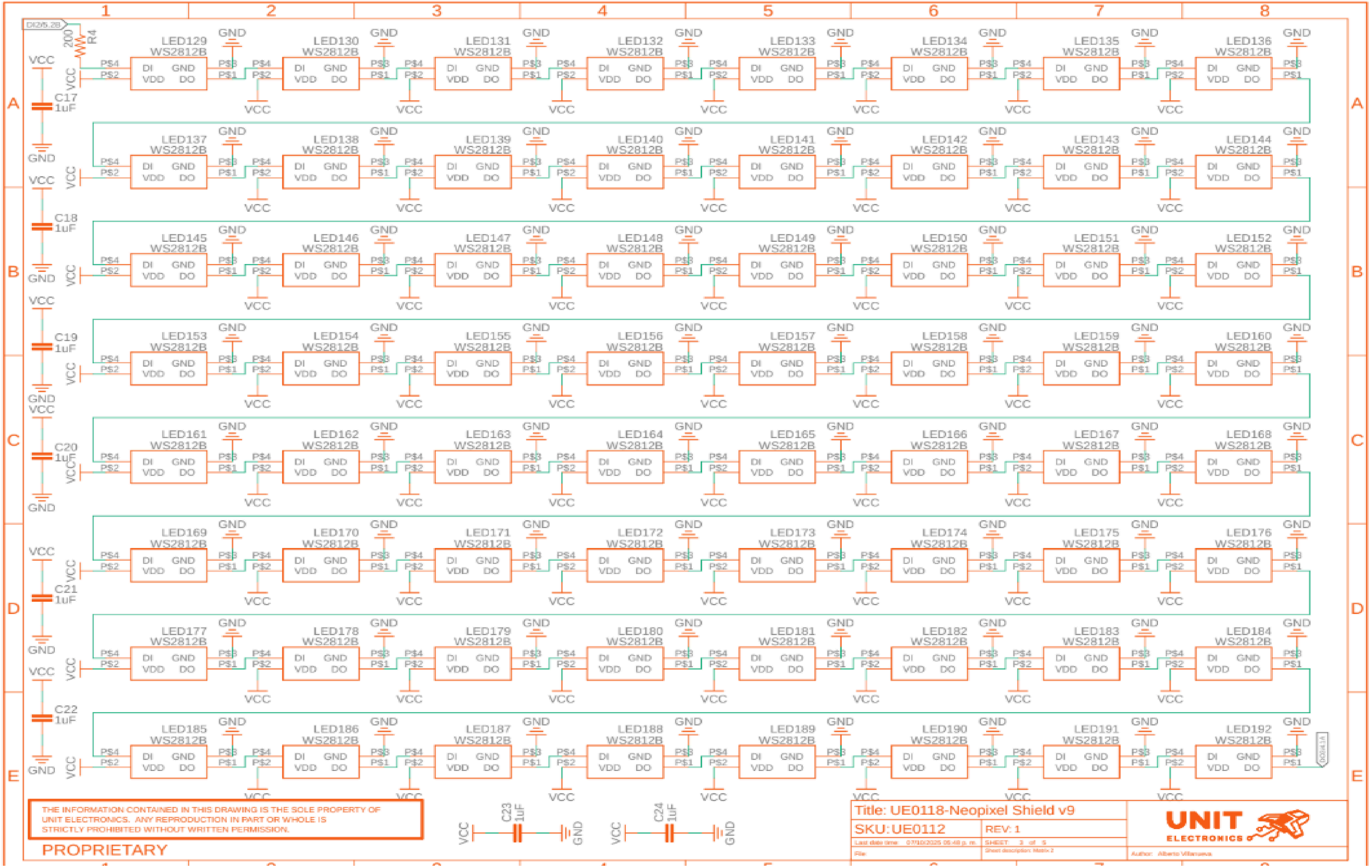
SHEET: 2 of 5
UNIT ELECTRONICS



LED129 WS2812B
LED130 WS2812B
LED131 WS2812B
LED132 WS2812B
LED133 WS2812B
LED134 WS2812B
LED135 WS2812B
LED136 WS2812B
LED185 WS2812B
LED192 WS2812B
C17 1uF
C18 1uF
C19 1uF
C20 1uF
C21 1uF
C22 1uF
C23 1uF
C24 1uF
VCC
GND
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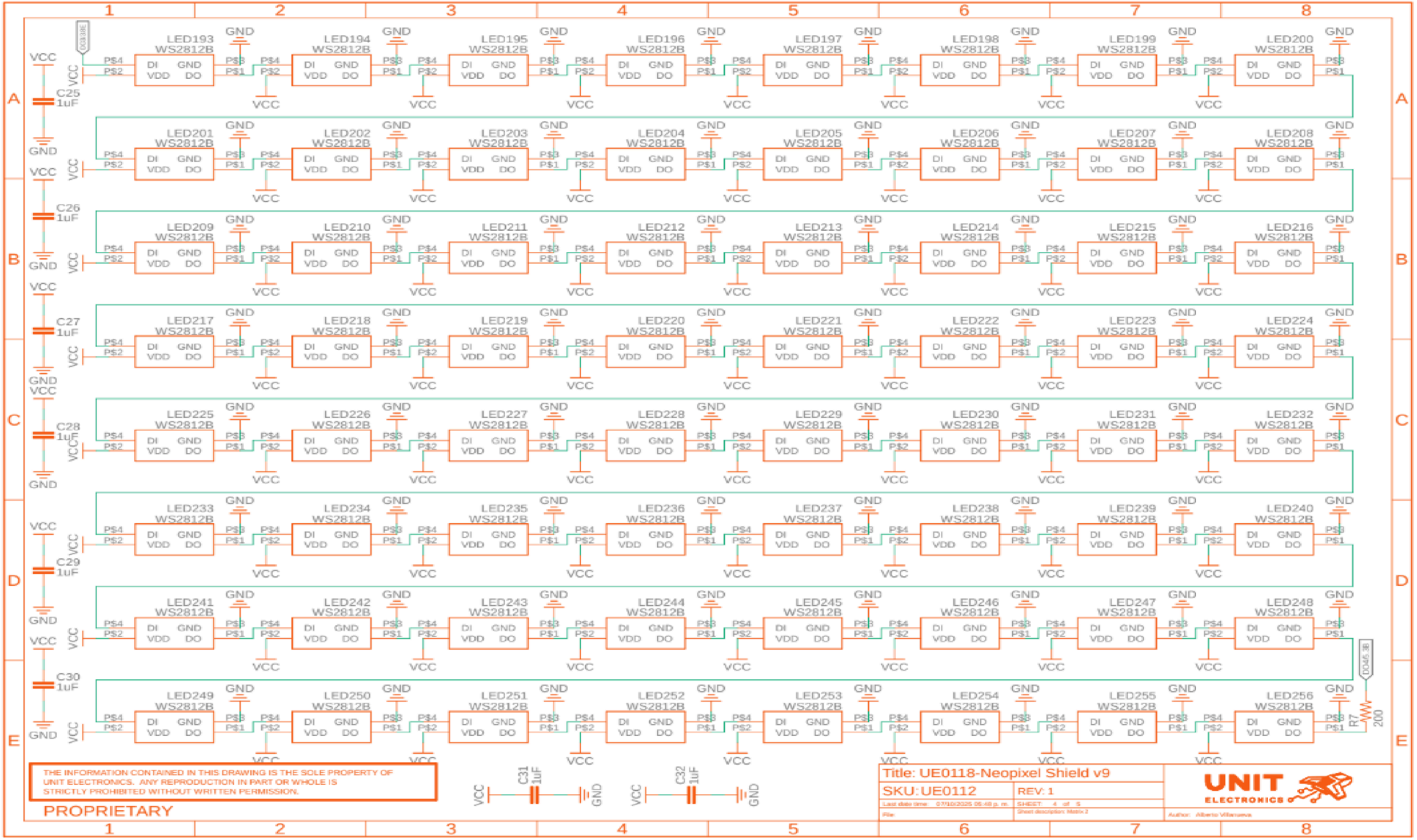
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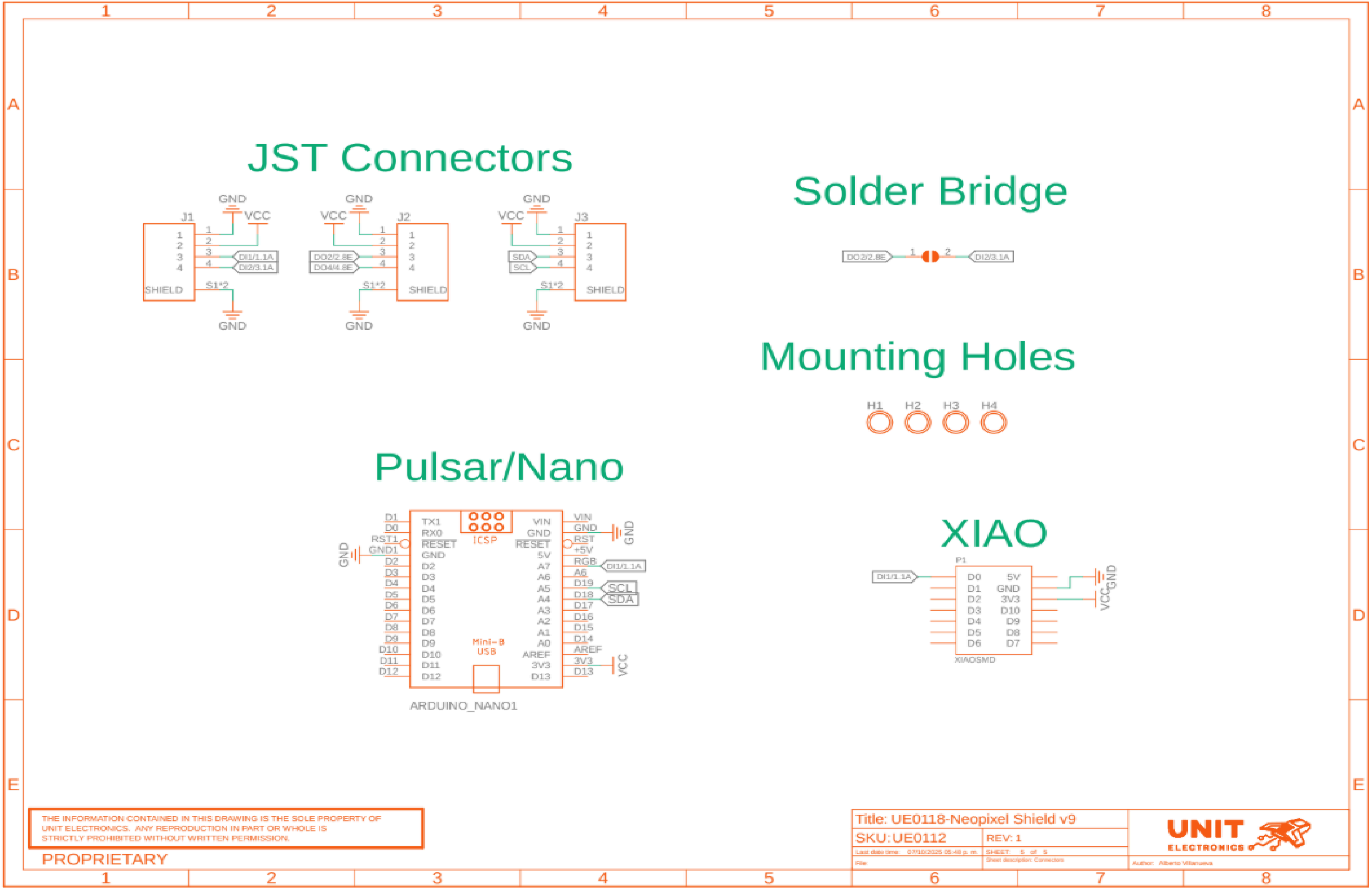
JST Connectors
Solder Bridge
Mounting Holes
Pulsar/Nano
XIAO
J1
J2
J3
GND
VCC
SHIELD
DI1/1.1A
DI2/3.1A
DO2/2.8E
DO4/4.8E
SDA
SCL
H1
H2
H3
H4
ARDUINO_NANO1
ICSP
Mini-B USB
P1
XIAOSMD
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